

Local Operators & Product Expansion

Local operators are those operators which are defined at a point $(z, \bar{z})$ on the complex world sheet. Sometimes, they are also referred to as fields / field operators. Eg: $\hat{\phi}(z, \bar{z})$, $\partial_\mu \hat{\phi}(z, \bar{z})$, $e^{\hat{\phi}(z, \bar{z})}$, etc are all called as local operators / fields.

Operator Product Expansion (OPE) is a statement about what happens as local operators approach each other. The idea is that two local operators inserted at nearby points can be approximated by a linear combination of a string of operators at of these points.

Let $\hat{O}_i(z_1, \bar{z}_1)$ & $\hat{O}_j(z_2, \bar{z}_2)$ denote two local operators at different points. The distance between these two points is got by $(u, \bar{u}) = (z_1 - z_2, \bar{z}_1 - \bar{z}_2)$.

Then, the product of these two local operators is then given by,

$$\hat{\mathcal{O}}_i(z_1, \bar{z}_1) \hat{\mathcal{O}}_j(z_2, \bar{z}_2) = \sum_k C_{ij}^k(z_1, \bar{z}_1) \hat{\mathcal{O}}_k(z_2, \bar{z}_2)$$

Here $C_{ij}^k(z_1, \bar{z}_1) \longrightarrow C_{ij}^k(z_1 - z_2, \bar{z}_1 - \bar{z}_2)$ are a set of functions that only depend only on the separation between the points the local operators depend on.

But the above should be thought of as operator eqⁿ. And by Hilbert Space formalism, it should be understood as statements that hold as the matrix elements of operators inside the time ordered symbol.

$$\langle \hat{\mathcal{O}}_i(z_1, \bar{z}_1) \hat{\mathcal{O}}_j(z_2, \bar{z}_2) \rangle = \sum_k C_{ij}^k(z_1, \bar{z}_1) \langle \hat{\mathcal{O}}_k(z_2, \bar{z}_2) \rangle$$

But still, we can include any other operator depending on a different complex point inside the angular brackets.

So, $\hat{\mathcal{O}}_i(z_1, \bar{z}_1) \hat{\mathcal{O}}_j(z_2, \bar{z}_2) = \sum_k C_{ij}^k(z, \bar{z}) \hat{\mathcal{O}}_k(z_2, \bar{z}_2)$
should always mean,

$$\langle T[\hat{\mathcal{O}}_i(z_1, \bar{z}_1) \hat{\mathcal{O}}_j(z_2, \bar{z}_2) \dots] \rangle = \sum_k C_{ij}^k(z, \bar{z}) \langle \hat{\mathcal{O}}_k(z_2, \bar{z}_2) \dots \rangle$$

'....' denotes any number of arbitrary operators. But they have to be at a distance larger compared to $|z_1 - z_2|$.

Inside the time ordering symbol the operators simply commute. But, for fermionic operators, as they are grassman's variable, they would pick up a -ve sign.

The OPE's have singular behaviour as $|z_1 - z_2| \rightarrow 0$. It is usually this, what we want to find out. It contains the same information of commutation relations.

In the series, the singular terms are kept first & then the non singular terms.

OPE's for the Closed String Scalar χ^μ Theory

We saw that the normal ordered product satisfies, $\partial_{z_1} \partial_{\bar{z}_1} : \hat{\chi}^\mu(z_1, \bar{z}_1) \hat{\chi}^\nu(z_2, \bar{z}_2) : = 0$

Then $: \hat{\chi}^\mu(z_1, \bar{z}_1) \hat{\chi}^\nu(z_2, \bar{z}_2) :$ is a Harmonic function of $(z_1, \bar{z}_1)$ which can be locally a sum of holomorphic and an anti-holomorphic function. So, it implies that there are no singular terms in its expansion as $z_1 \rightarrow z_2$.

Expand $\hat{\chi}^\mu(z_1, \bar{z}_1)$, about the point $(z_2, \bar{z}_2)$

$$\begin{aligned} \hat{\chi}^\mu(z_1, \bar{z}_1) &= \hat{\chi}^\mu(z_2, \bar{z}_2) + z_{12} \partial \hat{\chi}^\mu \Big|_{z_2, \bar{z}_2} \\ &+ \bar{z}_{12} \bar{\partial} \hat{\chi}^\mu \Big|_{z_2, \bar{z}_2} + \frac{z_{12}^2}{2} \partial^2 \hat{\chi}^\mu \Big|_{z_2, \bar{z}_2} + \frac{\bar{z}_{12}^2}{2} \bar{\partial}^2 \hat{\chi}^\mu \Big|_{z_2, \bar{z}_2} \\ &+ \frac{z_{12} \bar{z}_{12}}{2} (\partial \bar{\partial} \hat{\chi}^\mu) \Big|_{z_2, \bar{z}_2} + \dots \end{aligned}$$

Gross terms like $(\partial \bar{\partial} \hat{\chi}^\mu)$ will not contribute because of Eqⁿ of motion. (will show later)

Now, multiply by $\hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2)$

$$\begin{aligned} \hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) &= \hat{\chi}^\mu(\bar{z}_2, \bar{\bar{z}}_2) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) \\ &+ \bar{z}_{12} (\partial \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) + \bar{\bar{z}}_{12} (\bar{\partial} \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) \\ &+ \frac{\bar{z}_{12}^2}{2} (\partial^2 \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) + \frac{\bar{\bar{z}}_{12}^2}{2} (\bar{\partial}^2 \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) + \dots \end{aligned}$$

Then do normal ordering throughout,

$$\begin{aligned} :\hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2): &= :\hat{\chi}^\mu(\bar{z}_2, \bar{\bar{z}}_2) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2): \\ &+ \bar{z}_{12} :(\partial \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2): + \bar{\bar{z}}_{12} :(\bar{\partial} \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2): \\ &+ \frac{\bar{z}_{12}^2}{2} :(\partial^2 \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2): + \frac{\bar{\bar{z}}_{12}^2}{2} :(\bar{\partial}^2 \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2): \\ &+ \dots \end{aligned}$$

Note that $:\hat{\chi}^\mu(\bar{z}_2, \bar{\bar{z}}_2) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2):$ doesn't diverge as it is harmonic & non-singular.

Use $:\hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2):$

$$= \hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) + \frac{\alpha' \eta^{\mu\nu}}{2} \ln |\bar{z}_1 - \bar{z}_2|^2$$

$$\begin{aligned}
\rightarrow \hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) &= -\frac{\alpha' \eta^{\mu\nu}}{2} \ln |\bar{z}_1 - \bar{z}_2|^2 \\
&+ : \hat{\chi}^\mu \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) : + \bar{z}_{12} : (\partial \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) : \\
&+ \bar{\bar{z}}_{12} : (\bar{\partial} \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) : + \frac{\bar{z}_{12}^2}{2} : (\partial^2 \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) : \\
&+ \frac{\bar{\bar{z}}_{12}^2}{2} : (\bar{\partial}^2 \hat{\chi}^\mu) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) : + \dots
\end{aligned}$$

Thus we can include all orders & write

$$\begin{aligned}
\hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) &= -\frac{\alpha' \eta^{\mu\nu}}{2} \ln |\bar{z}_{12}|^2 + : \hat{\chi}^\mu \hat{\chi}^\nu : \\
&+ \sum_{k=1}^{\infty} \frac{1}{k!} \left[(\bar{z}_{12})^k : \hat{\chi}^\nu (\partial^k \hat{\chi}^\mu) : + (\bar{\bar{z}}_{12})^k : \hat{\chi}^\nu (\bar{\partial}^k \hat{\chi}^\mu) : \right]
\end{aligned}$$

Note that only the log term is singular as $\bar{z}_1 \rightarrow \bar{z}_2$, it is an essential singularity. All other terms are non singular.

$$\begin{aligned}
\rightarrow \hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) &\sim -\frac{\alpha' \eta^{\mu\nu}}{2} \ln |\bar{z}_{12}|^2 \\
&\quad \downarrow \\
&\text{Equal upto singular terms}
\end{aligned}$$

The normal ordering for more than two operators can be done recursively.

$$\text{Eg: } :\hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) \hat{\chi}^\rho(\bar{z}_3, \bar{\bar{z}}_3):$$

$$\begin{aligned} &= \hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) \hat{\chi}^\rho(\bar{z}_3, \bar{\bar{z}}_3) \\ &\quad + \frac{\alpha' \eta^{\mu\nu}}{2} \ln |\bar{z}_1 - \bar{z}_2|^2 \hat{\chi}^\rho(\bar{z}_3, \bar{\bar{z}}_3) \\ &\quad + \frac{\alpha' \eta^{\nu\rho}}{2} \ln |\bar{z}_2 - \bar{z}_3|^2 \hat{\chi}^\mu(\bar{z}_1, \bar{\bar{z}}_1) \\ &\quad + \frac{\alpha' \eta^{\rho\mu}}{2} \ln |\bar{z}_3 - \bar{z}_1|^2 \hat{\chi}^\nu(\bar{z}_2, \bar{\bar{z}}_2) \end{aligned}$$

$$\text{Let, } \hat{\chi}_i^\mu = \hat{\chi}^\mu(\bar{z}_i, \bar{\bar{z}}_i) \text{ \& } C_{ij}^{\mu\nu} = -\frac{\alpha' \eta^{\mu\nu}}{2} \ln |\bar{z}_i - \bar{z}_j|^2$$

$$\text{So, we get, } :\hat{\chi}_1^\mu \hat{\chi}_2^\nu \hat{\chi}_3^\rho \hat{\chi}_4^\gamma: = \hat{\chi}_1^\mu \hat{\chi}_2^\nu \hat{\chi}_3^\rho \hat{\chi}_4^\gamma$$

$$- C_{12}^{\mu\nu} \hat{\chi}_3^\rho \hat{\chi}_4^\gamma - C_{13}^{\mu\rho} \hat{\chi}_2^\nu \hat{\chi}_4^\gamma - C_{14}^{\mu\gamma} \hat{\chi}_3^\rho \hat{\chi}_2^\nu$$

$$- C_{23}^{\nu\rho} \hat{\chi}_1^\mu \hat{\chi}_4^\gamma - C_{24}^{\nu\gamma} \hat{\chi}_1^\mu \hat{\chi}_3^\rho - C_{34}^{\rho\gamma} \hat{\chi}_1^\mu \hat{\chi}_2^\nu$$

$$+ C_{12}^{\mu\nu} C_{34}^{\rho\gamma} + C_{13}^{\mu\rho} C_{24}^{\nu\gamma} + C_{14}^{\mu\gamma} C_{23}^{\nu\rho}$$

This can be generalized to many χ^μ 's

$$:\hat{\chi}_1^{\mu_1} \hat{\chi}_2^{\mu_2} \hat{\chi}_3^{\mu_3} \dots \hat{\chi}_n^{\mu_n}: = \hat{\chi}_1^{\mu_1} \hat{\chi}_2^{\mu_2} \hat{\chi}_3^{\mu_3} \dots \hat{\chi}_n^{\mu_n} + \sum \text{Sum of all possible Pairings}$$

Here, the sum runs over all possible ways of choosing one, two, or more pairs of operators & replacing each pair $\hat{\chi}_i^{\mu_i} \hat{\chi}_j^{\mu_j}$ by $C_{ij}^{\mu_i \mu_j}$. This is similar to Wick's theorem.

We can show that this quantity given by $:\hat{\chi}_1^{\mu_1} \hat{\chi}_2^{\mu_2} \hat{\chi}_3^{\mu_3} \dots \hat{\chi}_n^{\mu_n}:$ satisfies the harmonic eqⁿ

$$\partial_{z_i} \partial_{\bar{z}_i} (:\hat{\chi}_1^{\mu_1} \hat{\chi}_2^{\mu_2} \hat{\chi}_3^{\mu_3} \dots \hat{\chi}_n^{\mu_n}:) = 0$$

for all $i = 1, 2, 3, \dots, n$. & $z_{ij} \neq 0$. This is because, $\partial_{z_i} \partial_{\bar{z}_i} \hat{\chi}_i^{\mu_i} = 0$ (Eqⁿ of motion).

Then, $\partial_{z_i} \partial_{\bar{z}_i} C_{ij} \propto \partial(z_{ij}) = 0$, since $z_{ij} \neq 0$. So, all the terms in the expansion give 0.

The expansion of the normal ordering of a functional of χ can be written as,

$$:\mathcal{F}: = \text{Exp} \left[\frac{\alpha' \eta^{\mu\nu}}{4} \int d\bar{z} d\bar{z}' \ln|\bar{z} - \bar{z}'|^2 \frac{\partial}{\partial \hat{\chi}^\mu(\bar{z}, \bar{z})} \frac{\partial}{\partial \hat{\chi}^\nu(\bar{z}', \bar{z}')} \right] \mathcal{F}$$

Here the double derivative contracts each pair of fields & the exponential sums over any number of pairs with the factorial cancelling the number of ways the derivatives act.

$$\rightarrow :\mathcal{F}: = \mathcal{F} + \sum \text{Sum of all possible Pairings}$$

Observe that the above relations can be reversed.

$$\hat{\chi}_1^\mu \hat{\chi}_2^\nu = :\hat{\chi}_1^\mu \hat{\chi}_2^\nu: + C_{ij}^{\mu\nu}$$

$$\begin{aligned} \hat{\chi}_1^\mu \hat{\chi}_2^\nu \hat{\chi}_3^\rho = & :\hat{\chi}_1^\mu \hat{\chi}_2^\nu \hat{\chi}_3^\rho: + C_{12}^{\mu\nu} :\hat{\chi}_3^\rho: + C_{23}^{\nu\rho} :\hat{\chi}_1^\mu: \\ & + C_{31}^{\rho\mu} :\hat{\chi}_2^\nu: \end{aligned}$$

$$\begin{aligned}
\hat{\chi}_1^\mu \hat{\chi}_2^\nu \hat{\chi}_3^\rho \hat{\chi}_4^\gamma &= :\hat{\chi}_1^\mu \hat{\chi}_2^\nu \hat{\chi}_3^\rho \hat{\chi}_4^\gamma: + C_{12}^{\mu\nu} : \hat{\chi}_3^\rho \hat{\chi}_4^\gamma : \\
&+ C_{13}^{\mu\rho} : \hat{\chi}_2^\nu \hat{\chi}_4^\gamma : + C_{14}^{\mu\gamma} : \hat{\chi}_3^\rho \hat{\chi}_2^\nu : + C_{23}^{\nu\rho} : \hat{\chi}_1^\mu \hat{\chi}_4^\gamma : \\
&+ C_{24}^{\nu\gamma} : \hat{\chi}_1^\mu \hat{\chi}_3^\rho : + C_{34}^{\rho\gamma} : \hat{\chi}_1^\mu \hat{\chi}_2^\nu : + C_{12}^{\mu\nu} C_{34}^{\rho\gamma} \\
&+ C_{13}^{\mu\rho} C_{24}^{\nu\gamma} + C_{14}^{\mu\gamma} C_{23}^{\nu\rho}
\end{aligned}$$

Thus we can write,

$$\mathcal{F} = \text{Exp} \left[-\frac{\alpha' \eta^{\mu\nu}}{4} \int d^2\bar{z} d^2\bar{z}' \ln|\bar{z} - \bar{z}'|^2 \frac{\partial}{\partial \hat{\chi}^\mu(\bar{z}, \bar{z})} \frac{\partial}{\partial \hat{\chi}^\nu(\bar{z}', \bar{z}')} \right] : \mathcal{F} :$$

$$\rightarrow \mathcal{F} = : \underbrace{\left[\mathcal{F} + \sum (\text{All possible Contractions}) \right]}_{\text{Wick's Theorem}} :$$

Also OPE for any Pair of operator functionals can be generated as follows.

$$:\mathcal{F}\mathcal{G}: = \text{Exp} \left[\frac{\alpha' \eta^{\mu\nu}}{2} \int d^2\bar{z} d^2\bar{z}' \ln|\bar{z} - \bar{z}'|^2 \frac{\partial}{\partial \hat{\chi}_F^\mu(\bar{z})} \frac{\partial}{\partial \hat{\chi}_G^\nu(\bar{z}')} \right] : \mathcal{F} : : \mathcal{G} :$$

$\downarrow$ acts on $\mathcal{F}$ $\downarrow$ acts on $\mathcal{G}$.

$$\rightarrow :F::G: = :FG: + \sum_{\text{all}} \text{Cross-Contractions}$$

Eg: We shall expand $: \partial_1 \hat{X}_1^\mu \partial_1 \hat{X}_1^\nu : : \partial_2 \hat{X}_2^\alpha \partial_2 \hat{X}_2^\beta :$
 where $\partial_i \equiv \partial_{z_i}$. The contraction b/w
 two derivatives is given by,

$$\begin{aligned} \overbrace{: \partial_{\bar{z}_1} \hat{X}_1^\mu \partial_{\bar{z}_2} \hat{X}_2^\alpha :} &= \partial_{\bar{z}_1} \partial_{\bar{z}_2} : \overbrace{\hat{X}_1^\mu \hat{X}_2^\alpha} : \sim \partial_{\bar{z}_1} \partial_{\bar{z}_2} (C_{12}^{\mu\alpha}) \\ &= \partial_{\bar{z}_1} \partial_{\bar{z}_2} \left(-\frac{\alpha'}{2} \eta^{\mu\alpha} \ln |z_1 - z_2|^2 \right) = -\frac{\alpha'}{2} \eta^{\mu\alpha} \partial_{\bar{z}_1} \partial_{\bar{z}_2} \ln |z_1 - z_2|^2 \\ &= -\frac{\alpha'}{2} \eta^{\mu\alpha} \partial_{\bar{z}_1} \partial_{\bar{z}_2} \left[\ln(z_1 - \bar{z}_2) + \underbrace{\ln(\bar{z}_1 - z_2)}_{\text{(no contribution)}} \right] \\ &= -\frac{\alpha'}{2} \eta^{\mu\alpha} \partial_{\bar{z}_1} \left(\frac{-1}{z_1 - \bar{z}_2} \right) = -\frac{\alpha'}{2} \eta^{\mu\alpha} \frac{1}{(z_1 - \bar{z}_2)^2} \\ \rightarrow \underline{\underline{: \partial_{\bar{z}_1} \hat{X}_1^\mu \partial_{\bar{z}_2} \hat{X}_2^\alpha : \sim -\frac{\alpha'}{2} \eta^{\mu\alpha} \frac{1}{(z_1 - \bar{z}_2)^2}}}} \end{aligned}$$

So we can write,

$$\overbrace{: \partial \hat{X}_1^\mu \partial \hat{X}_2^\alpha :} : \overbrace{: \partial \hat{X}_1^\nu \partial \hat{X}_2^\beta :} \sim \frac{\alpha'^2}{4} \eta^{\mu\alpha} \eta^{\nu\beta} \frac{1}{(z_1 - z_2)^4}$$

Now we can expand the above term.

$$:\partial_1 \hat{\chi}_1^\mu \partial_1 \hat{\chi}_1^\nu :: \partial_2 \hat{\chi}_2^\alpha \partial_2 \hat{\chi}_2^\beta : = :\partial_1 \hat{\chi}_1^\mu \partial_1 \hat{\chi}_1^\nu \partial_2 \hat{\chi}_2^\alpha \partial_2 \hat{\chi}_2^\beta :$$

$$- \frac{\alpha'}{2} \frac{1}{(\bar{z}_1 - \bar{z}_2)^2} \left(\eta^{\mu\alpha} : \partial_1 \hat{\chi}_1^\nu \partial_2 \hat{\chi}_2^\beta : + \eta^{\mu\beta} : \partial_1 \hat{\chi}_1^\nu \partial_2 \hat{\chi}_2^\alpha : \right. \\ \left. + \eta^{\nu\alpha} : \partial_1 \hat{\chi}_1^\mu \partial_2 \hat{\chi}_2^\beta : + \eta^{\nu\beta} : \partial_1 \hat{\chi}_1^\mu \partial_2 \hat{\chi}_2^\alpha : \right) \\ + \frac{\alpha'^2}{4} \frac{1}{(\bar{z}_1 - \bar{z}_2)^4} \left(\eta^{\mu\alpha} \eta^{\nu\beta} + \eta^{\mu\beta} \eta^{\nu\alpha} \right)$$

$$\rightarrow \eta_{\mu\nu} \eta_{\alpha\beta} : \partial_1 \hat{\chi}_1^\mu \partial_1 \hat{\chi}_1^\nu :: \partial_2 \hat{\chi}_2^\alpha \partial_2 \hat{\chi}_2^\beta :$$

$$= \eta_{\mu\nu} \eta_{\alpha\beta} : \partial_1 \hat{\chi}_1^\mu \partial_1 \hat{\chi}_1^\nu \partial_2 \hat{\chi}_2^\alpha \partial_2 \hat{\chi}_2^\beta :$$

$$- \frac{\alpha'}{2} \frac{\eta_{\mu\nu} \eta_{\alpha\beta}}{(\bar{z}_1 - \bar{z}_2)^2} \left(\eta^{\mu\alpha} : \partial_1 \hat{\chi}_1^\nu \partial_2 \hat{\chi}_2^\beta : + \eta^{\mu\beta} : \partial_1 \hat{\chi}_1^\nu \partial_2 \hat{\chi}_2^\alpha : \right. \\ \left. + \eta^{\nu\alpha} : \partial_1 \hat{\chi}_1^\mu \partial_2 \hat{\chi}_2^\beta : + \eta^{\nu\beta} : \partial_1 \hat{\chi}_1^\mu \partial_2 \hat{\chi}_2^\alpha : \right) \\ + \frac{\alpha'^2}{4} \frac{\eta_{\mu\nu} \eta_{\alpha\beta}}{(\bar{z}_1 - \bar{z}_2)^4} \left(\eta^{\mu\alpha} \eta^{\nu\beta} + \eta^{\mu\beta} \eta^{\nu\alpha} \right)$$

$$= : \partial_1 \hat{\chi}_1^\mu \partial_1 \hat{\chi}_{1\mu} \partial_2 \hat{\chi}_2^\alpha \partial_2 \hat{\chi}_{2\alpha} :$$

$$- \frac{\alpha'}{2(\bar{z}_1 - \bar{z}_2)^2} \left(\eta_{\beta\nu} : \partial_1 \hat{\chi}_1^\nu \partial_2 \hat{\chi}_2^\beta : + \eta_{\nu\alpha} : \partial_1 \hat{\chi}_1^\nu \partial_2 \hat{\chi}_2^\alpha : \right. \\ \left. + \eta_{\mu\beta} : \partial_1 \hat{\chi}_1^\mu \partial_2 \hat{\chi}_2^\beta : + \eta_{\mu\alpha} : \partial_1 \hat{\chi}_1^\mu \partial_2 \hat{\chi}_2^\alpha : \right) \\ + \frac{\alpha'^2}{4} \frac{1}{(\bar{z}_1 - \bar{z}_2)^4} \left(\eta_{\mu\nu} \eta^{\mu\nu} + \eta^{\mu\nu} \eta_{\mu\nu} \right)$$

$$\rightarrow : \partial_1 \hat{\chi}_1^\mu \partial_1 \hat{\chi}_{1\mu} : : \partial_2 \hat{\chi}_2^\nu \partial_2 \hat{\chi}_{2\nu} :$$

$$= : \partial_1 \hat{\chi}_1^\mu \partial_1 \hat{\chi}_{1\mu} \partial_2 \hat{\chi}_2^\nu \partial_2 \hat{\chi}_{2\nu} : - \frac{4\alpha'}{2(\bar{z}_1 - \bar{z}_2)^2} : \partial_1 \hat{\chi}_1^\mu \partial_2 \hat{\chi}_{2\mu} : \\ + 2 \delta^\mu{}_\mu \frac{\alpha'^2}{4} \frac{1}{(\bar{z}_1 - \bar{z}_2)^4}$$

$$\rightarrow : \partial_1 \hat{\chi}_1^\mu \partial_1 \hat{\chi}_{1\mu} : : \partial_2 \hat{\chi}_2^\nu \partial_2 \hat{\chi}_{2\nu} :$$

$$= \frac{D\alpha'^2}{2} \frac{1}{(\bar{z}_1 - \bar{z}_2)^4} - \frac{2\alpha'}{(\bar{z}_1 - \bar{z}_2)^2} : \partial \hat{\chi}_1^\mu \partial \hat{\chi}_{2\mu} :$$

$$+ : \partial \hat{\chi}_1^\mu \partial \hat{\chi}_{1\mu} \partial \hat{\chi}_2^\nu \partial \hat{\chi}_{2\nu} :$$

Consider $\partial_1 \hat{\chi}_1^\mu = \left(\partial \hat{\chi}^\mu \right)_{\bar{z}_2} + (\bar{z}_1 - \bar{z}_2) \left(\partial^2 \hat{\chi}^\mu \right)_{\bar{z}_2} \\ + \frac{(\bar{z}_1 - \bar{z}_2)^2}{2!} \left(\partial^3 \hat{\chi}^\mu \right)_{\bar{z}_2} + \dots$

$$: \partial_1 \hat{\chi}_1^\mu \partial \hat{\chi}_{2\mu} : = : \partial \hat{\chi}_2^\mu \partial \hat{\chi}_{2\mu} : + (\bar{z}_1 - \bar{z}_2) : \left(\partial^2 \hat{\chi}_2^\mu \right) \partial \hat{\chi}_{2\mu} : \\ + \frac{(\bar{z}_1 - \bar{z}_2)^2}{2!} : \left(\partial^3 \hat{\chi}_2^\mu \right) \partial \hat{\chi}_{2\mu} : + \dots$$

(Higher terms can be neglected.)

$$\begin{aligned}
& \rightarrow : \partial_1 \hat{X}_1^\mu \partial_1 \hat{X}_{1\mu} : : \partial_2 \hat{X}_2^\nu \partial_2 \hat{X}_{2\nu} : \\
& = \frac{D\alpha'^2}{2} \frac{1}{(\bar{z}_1 - \bar{z}_2)^4} - \frac{2\alpha'}{(\bar{z}_1 - \bar{z}_2)^2} : \partial \hat{X}_2^\mu \partial \hat{X}_{2\mu} : \\
& \quad - \frac{2\alpha'}{(\bar{z}_1 - \bar{z}_2)} : (\partial^2 \hat{X}_2^\mu) \partial \hat{X}_{2\mu} : - \alpha' : (\partial^3 \hat{X}_2^\mu) \partial \hat{X}_{2\mu} : + \dots \\
& \quad + : \partial_2 \hat{X}^\mu \partial_2 \hat{X}_\mu \partial_2 \hat{X}^\nu \partial_2 \hat{X}_{2\nu} : + \dots
\end{aligned}$$

Thus neglecting the non-singular term, we get,

$$\begin{aligned}
& : \partial_1 \hat{X}_1^\mu \partial_1 \hat{X}_{1\mu} : : \partial_2 \hat{X}_2^\nu \partial_2 \hat{X}_{2\nu} : \\
& \sim \frac{D\alpha'^2}{2} \frac{1}{(\bar{z}_1 - \bar{z}_2)^4} - \frac{2\alpha'}{(\bar{z}_1 - \bar{z}_2)^2} : \partial \hat{X}_2^\mu \partial \hat{X}_{2\mu} : \\
& \quad - \frac{2\alpha'}{(\bar{z}_1 - \bar{z}_2)} : (\partial^2 \hat{X}_2^\mu) \partial \hat{X}_{2\mu} :
\end{aligned}$$

This is the operator product expansion & we have only included the singular terms & neglected the non-singular term.

Another example is,

$$\mathcal{F} = e^{ip_\mu \hat{X}^\mu(z_1, \bar{z}_1)}, \quad \mathcal{G} = e^{iq_\nu \hat{X}^\nu(z_2, \bar{z}_2)}$$

$$\frac{\partial}{\partial \hat{X}_F^\mu(z)} \mathcal{F} = ip_\mu \mathcal{F}, \quad \frac{\partial}{\partial \hat{X}_G^\nu(z')} \mathcal{G} = iq_\nu \mathcal{G}$$

$$\text{Thus } \frac{\partial}{\partial \hat{X}_F^\mu(z)} \rightarrow ip_\mu \quad \& \quad \frac{\partial}{\partial \hat{X}_G^\nu(z')} \rightarrow iq_\nu$$

$$\rightarrow \text{Exp} \left[-\frac{\alpha'}{2} \eta^{\mu\nu} \int d^2z d^2z' \ln|z-z'|^2 \frac{\partial}{\partial \hat{X}_F^\mu(z)} \frac{\partial}{\partial \hat{X}_G^\nu(z')} \right]$$

$$\hookrightarrow \text{Exp} \left[-\frac{\alpha'}{2} \eta^{\mu\nu} \ln|z_1-z_2|^2 ip_\mu iq_\nu \right]$$

$$= e^{\left[\alpha' \ln|z_1-z_2| p^\mu q_\mu \right]} = |z_1-z_2|^{\alpha' p^\mu q_\mu}$$

Thus we can write,

$$:e^{ip_\mu \hat{X}_1^\mu}::e^{iq_\nu \hat{X}_2^\nu}: = |z_1-z_2|^{\alpha' p^\mu q_\mu} :e^{ip_\mu \hat{X}_1^\mu} e^{iq_\nu \hat{X}_2^\nu}:$$

Now we shall expand $:e^{ip_\mu \hat{\chi}_1^\mu} e^{iq_\mu \hat{\chi}_2^\mu}: to get OPE. Now, Taylor expand,$

$$e^{ip_\mu \hat{\chi}_1^\mu} = e^{ip_\mu \hat{\chi}_2^\mu} + z_{12} i p_\mu (\partial \hat{\chi}_2^\mu) e^{ip_\mu \hat{\chi}_2^\mu} + \bar{z}_{12} i p_\mu (\bar{\partial} \hat{\chi}_2^\mu) e^{ip_\mu \hat{\chi}_2^\mu} + \dots$$

$$= e^{ip_\mu \hat{\chi}_2^\mu} \left(1 + i p_\mu \left[z_{12} (\partial \hat{\chi}_2^\mu) + \bar{z}_{12} (\bar{\partial} \hat{\chi}_2^\mu) \right] + \dots \right)$$

$$\rightarrow :e^{ip_\mu \hat{\chi}_1^\mu} e^{iq_\mu \hat{\chi}_2^\mu}:$$

$$=:e^{i(p_\mu + q_\mu) \hat{\chi}_2^\mu} \left(1 + i p_\mu \left[z_{12} (\partial \hat{\chi}_2^\mu) + \bar{z}_{12} (\bar{\partial} \hat{\chi}_2^\mu) \right] + \dots \right):$$

$$:e^{ip_\mu \hat{\chi}_1^\mu} ::e^{iq_\mu \hat{\chi}_2^\mu}: = |z_1 - z_2|^{\alpha' p^\mu q_\mu}$$

$$\times :e^{i(p_\mu + q_\mu) \hat{\chi}_2^\mu} \left(1 + i p_\mu \left[z_{12} (\partial \hat{\chi}_2^\mu) + \bar{z}_{12} (\bar{\partial} \hat{\chi}_2^\mu) \right] + \dots \right):$$

Take $(z_1, \bar{z}_1) \rightarrow (z, \bar{z}) \neq (z_2, \bar{z}_2) \rightarrow (0, 0)$

$$:e^{ip_\mu \hat{\chi}_1^\mu} ::e^{iq_\mu \hat{\chi}_2^\mu}: = |z|^{\alpha' p^\mu q_\mu} :e^{i(p_\mu + q_\mu) \hat{\chi}_2^\mu} (1 + O(z, \bar{z})):$$

$O(z, \bar{z}) \rightarrow \text{Some local operator}$

To find OPE $\hat{X}^\mu(z_1, \bar{z}_1) : e^{ik_\nu \hat{X}^\nu(z_2, \bar{z}_2)} :$

$$\hat{X}^\mu(z_1, \bar{z}_1) : e^{ik_\nu \hat{X}^\nu(z_2, \bar{z}_2)} :$$

$$= \hat{X}^\mu(z_1, \bar{z}_1) \sum_{n=0}^{\infty} \frac{i^n : (k_\nu \hat{X}^\nu(z_2, \bar{z}_2))^n :}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{i^n}{n!} k_{\nu_1} k_{\nu_2} \dots k_{\nu_n} \underbrace{\hat{X}^\mu(z_1, \bar{z}_1) : \hat{X}^{\nu_1} \hat{X}^{\nu_2} \dots \hat{X}^{\nu_n}(z_2, \bar{z}_2) :}_{n\text{-possible contractions}}$$

$$\hat{X}^\mu(z_1, \bar{z}_1) \hat{X}^{\nu_1}(z_2, \bar{z}_2) \sim -\frac{\alpha'}{2} \eta^{\mu\nu_1} \ln|z_1 - z_2|^2$$

$$\Rightarrow \hat{X}^\mu(z_1, \bar{z}_1) : e^{ik_\nu \hat{X}^\nu(z_2, \bar{z}_2)} :$$

$$\sim \sum_{n=0}^{\infty} \frac{i^n}{n!} k_{\nu_1} k_{\nu_2} \dots k_{\nu_n} \left(-\frac{\alpha'}{2} \ln|z_1 - z_2|^2 \right)$$

$$\left(\eta^{\mu\nu_1} : \hat{X}^{\nu_2} \dots \hat{X}^{\nu_n}(z_2, \bar{z}_2) : \right.$$

$$+ \eta^{\mu\nu_2} : \hat{X}^{\nu_1} \hat{X}^{\nu_3} \dots \hat{X}^{\nu_n}(z_2, \bar{z}_2) :$$

$$+ \dots + \eta^{\mu\nu_n} : \hat{X}^{\nu_1} \hat{X}^{\nu_2} \dots \hat{X}^{\nu_{n-1}}(z_2, \bar{z}_2) :)$$

$\underbrace{\hspace{10em}}_{n \text{ terms all equal}}$

$$= \left(-\frac{\alpha' \ln |z_1 - z_2|^2}{2} \right) \sum_{n=0}^{\infty} \frac{i^n k^\mu n}{n!} : (k_\nu \hat{\chi}^\nu(z_2, \bar{z}_2))^{n-1} :$$

$$= \left(-\frac{\alpha' \ln |z_1 - z_2|^2}{2} \right) i k^\mu \underbrace{\sum_{n=1}^{\infty} \frac{i^{n-1}}{(n-1)!} : (k_\nu \hat{\chi}^\nu(z_2, \bar{z}_2))^{n-1} :}_{\text{Exponential}} :$$

$$= -\frac{i k^\mu \alpha' \ln |z_{12}|^2}{2} : e^{i k_\nu \hat{\chi}^\nu(z_2, \bar{z}_2)} :$$

$$\hat{\chi}^\mu(z_1, \bar{z}_1) : e^{i k_\nu \hat{\chi}^\nu(z_2, \bar{z}_2)} :$$

$$\sim -\frac{i k^\mu \alpha' \ln |z_{12}|^2}{2} : e^{i k_\nu \hat{\chi}^\nu(z_2, \bar{z}_2)} :$$

Similarly,

$$\partial_{z_1} \hat{\chi}^\mu(z_1, \bar{z}_1) : e^{i k_\nu \hat{\chi}^\nu(z_2, \bar{z}_2)} :$$

$$\sim \frac{-i k^\mu \alpha'}{2(z_1 - z_2)} : e^{i k_\nu \hat{\chi}^\nu(z_2, \bar{z}_2)} :$$